

Winding Ambiguity in ω Bridging

Weekly Update — 10 March 2026

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Where We Left Off

Full graph pipeline: 50 candidates $\times$ 3 brightness peaks, shortest-path graph search.

Stage	Result
Candidate generation	1,400–2,987 hits/peak from 1M random samples
Bridge feasibility	98.5% feasible — 3 free ω DOF connects almost any pair
Lo-fi intermediate scoring	Truth bridge indistinguishable from random
Hi-fi rescoring	Truth ranked #13,979 / 121,326

Your prescription: (1) search for the *minimum*- $|\omega|$ bridge; (2) explore the winding family; (3) score by trough brightness rather than intermediate epochs.

Root Cause: Winding Degeneracy

A fast tumbler bridges two peaks over $\Delta t = 554\text{s}$ at $|\boldsymbol{\omega}| \approx 2.08^\circ/\text{s}$ — roughly **3 full rotations**.

The axis-angle estimate assumes the *shortest arc* — zero extra loops.

	True $\boldsymbol{\omega}$	Axis-angle bridge
Magnitude	2.08 °/s	0.32 °/s
Winding number	~3	~0
Arrival mismatch	—	0.0° ✓
Intermediate trajectory	Correct	Completely wrong

The solver finds a trajectory 8× too slow — yet arrives at exactly the right attitude.

Arrival matching alone cannot distinguish winding numbers.

Why This Killed the Graph Pipeline

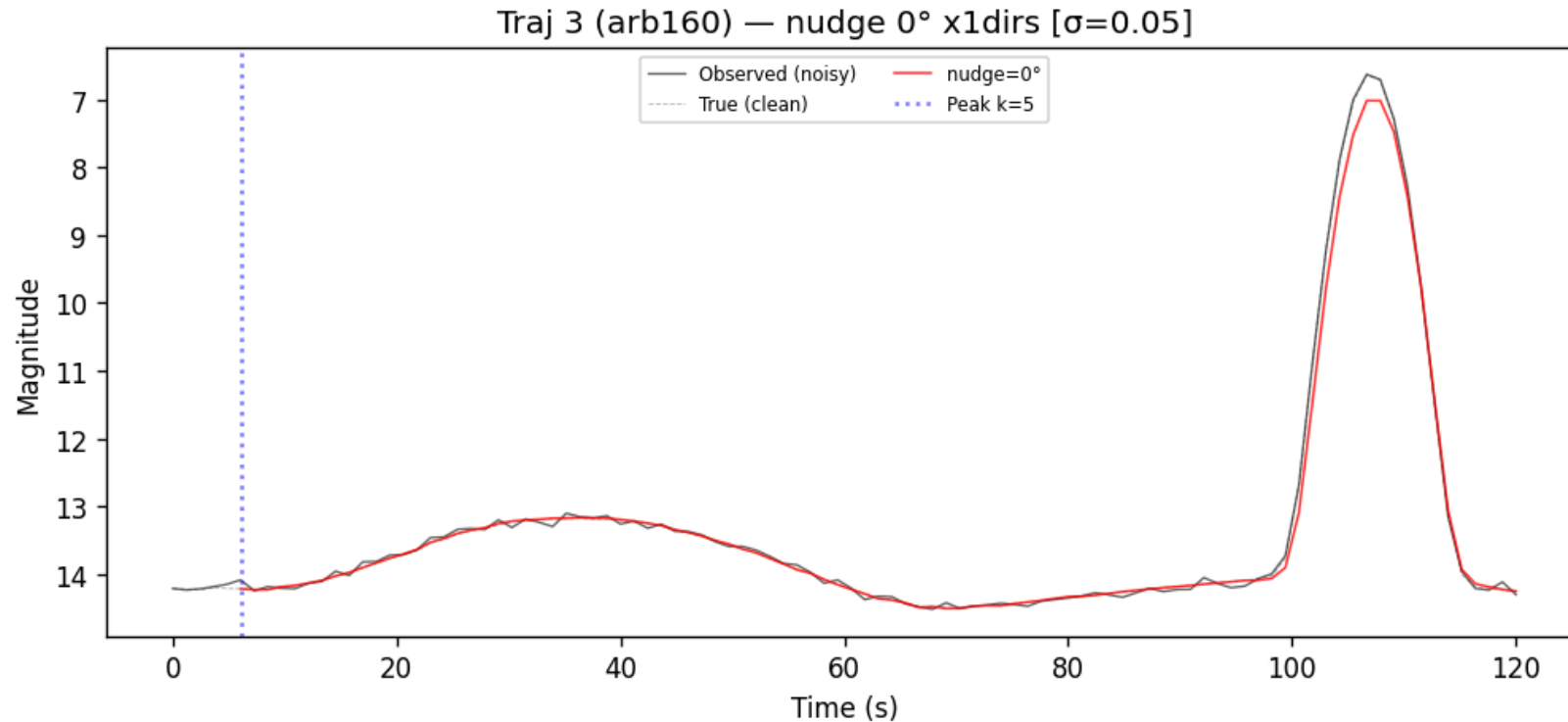
At intermediate epochs, all feasible bridges predict *some* attitude — but if the winding number is wrong, that attitude is completely wrong. No scoring signal survives.

- Lo-fi intermediate scoring → scores uniform across all paths
- Hi-fi rescoring → same problem: the bridge ω itself has the wrong winding

The fix must happen inside the bridge optimisation, not in the scoring step.

Omega Recovery: Exact Attitudes Work Perfectly

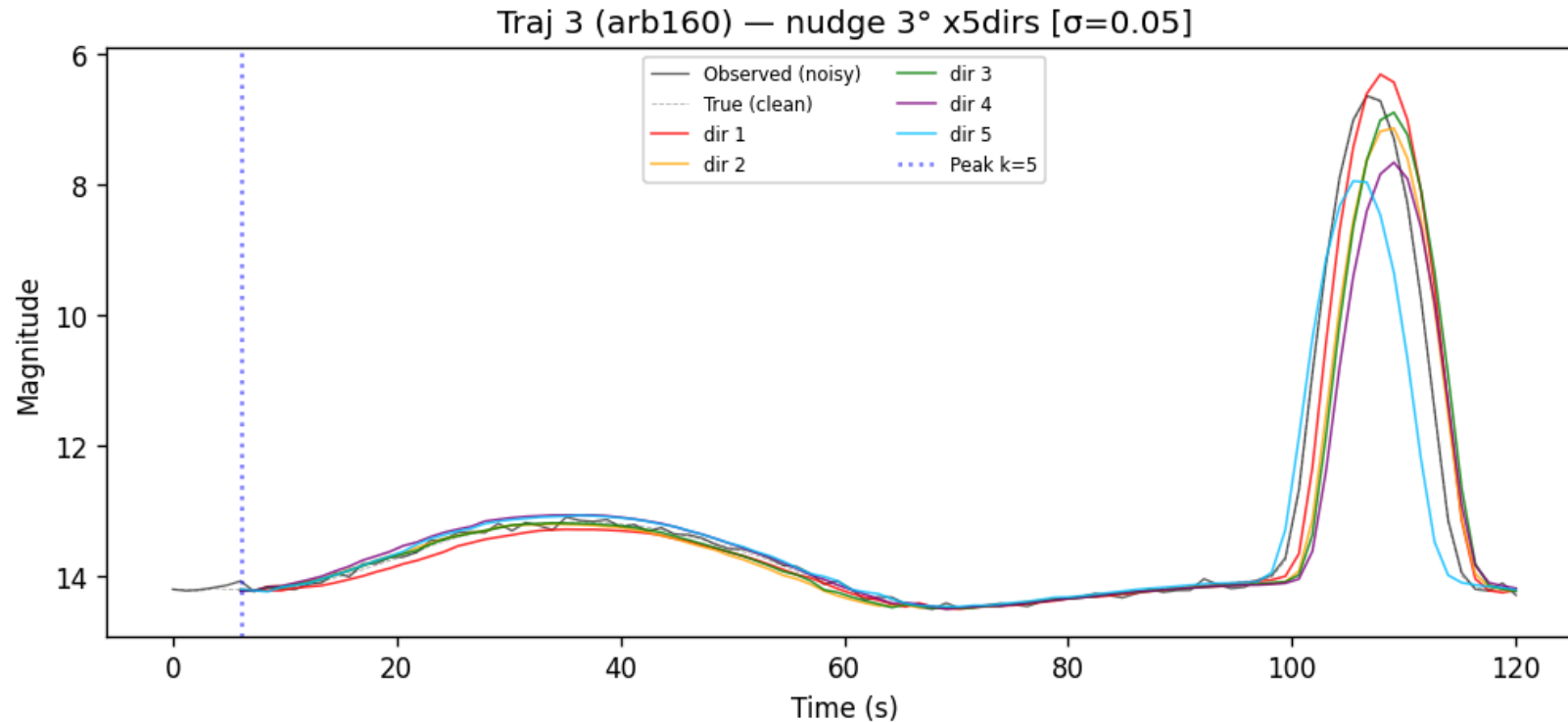
Given exact quaternion endpoints, recovering ω is trivial and cheap.



Metric	Value
Cost per bridge solve	~160 ms (L-BFGS-B, 44 evals)
Omega error at zero nudge	< 0.000001 °/s

Hi-fi: Attitude Error Is Anisotropic

3° nudge in 5 random directions, hi-fi reconstruction from the recovered ω :

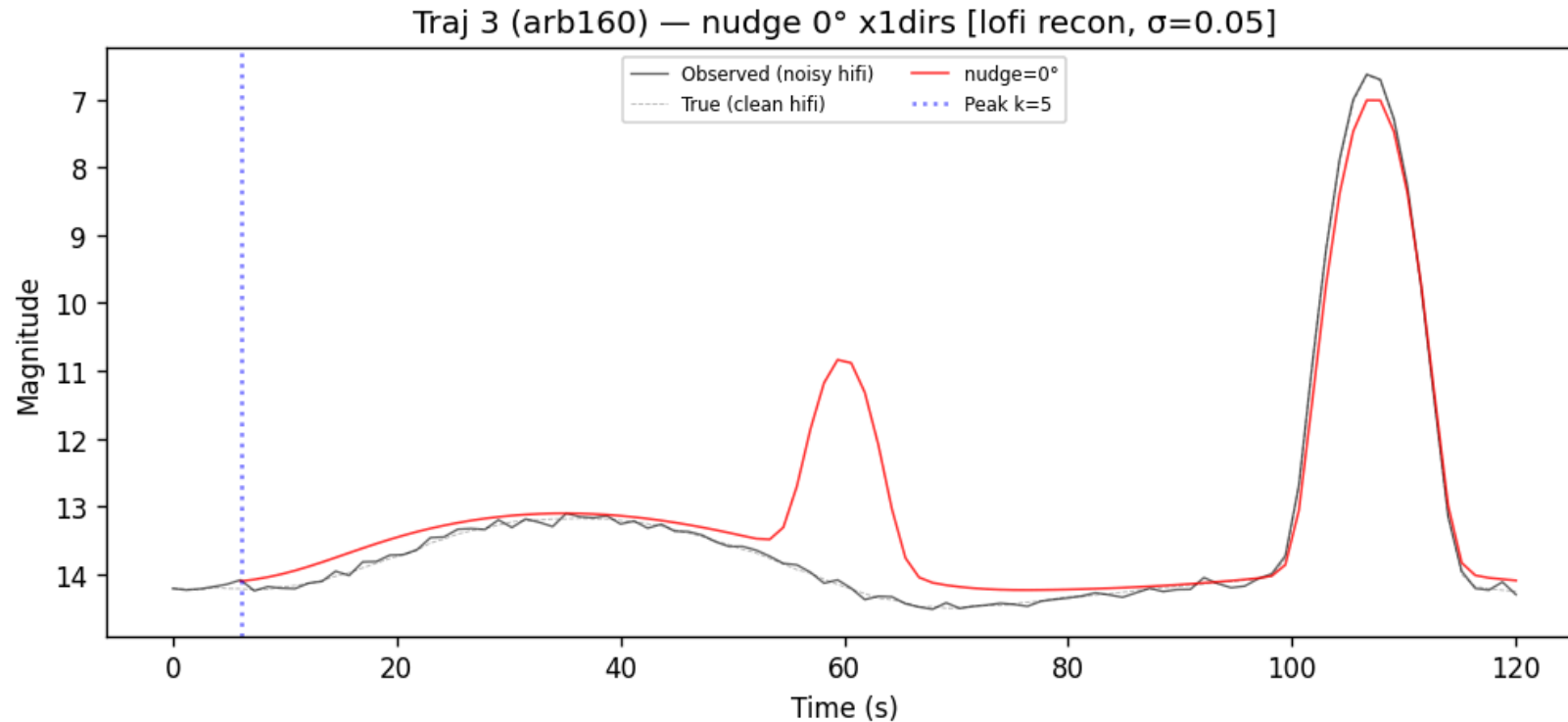


On average: ~ 0.025 °/s ω error per degree of attitude error.

But **direction matters** — at the glint peak ($t \approx 107$ s), some directions produce near-perfect reconstructions while others shift the peak by >2 mag.

Lo-fi: Phantom Peaks From Self-Shadowing

Lo-fi reconstruction with **zero attitude error** — the omega is perfectly recovered:

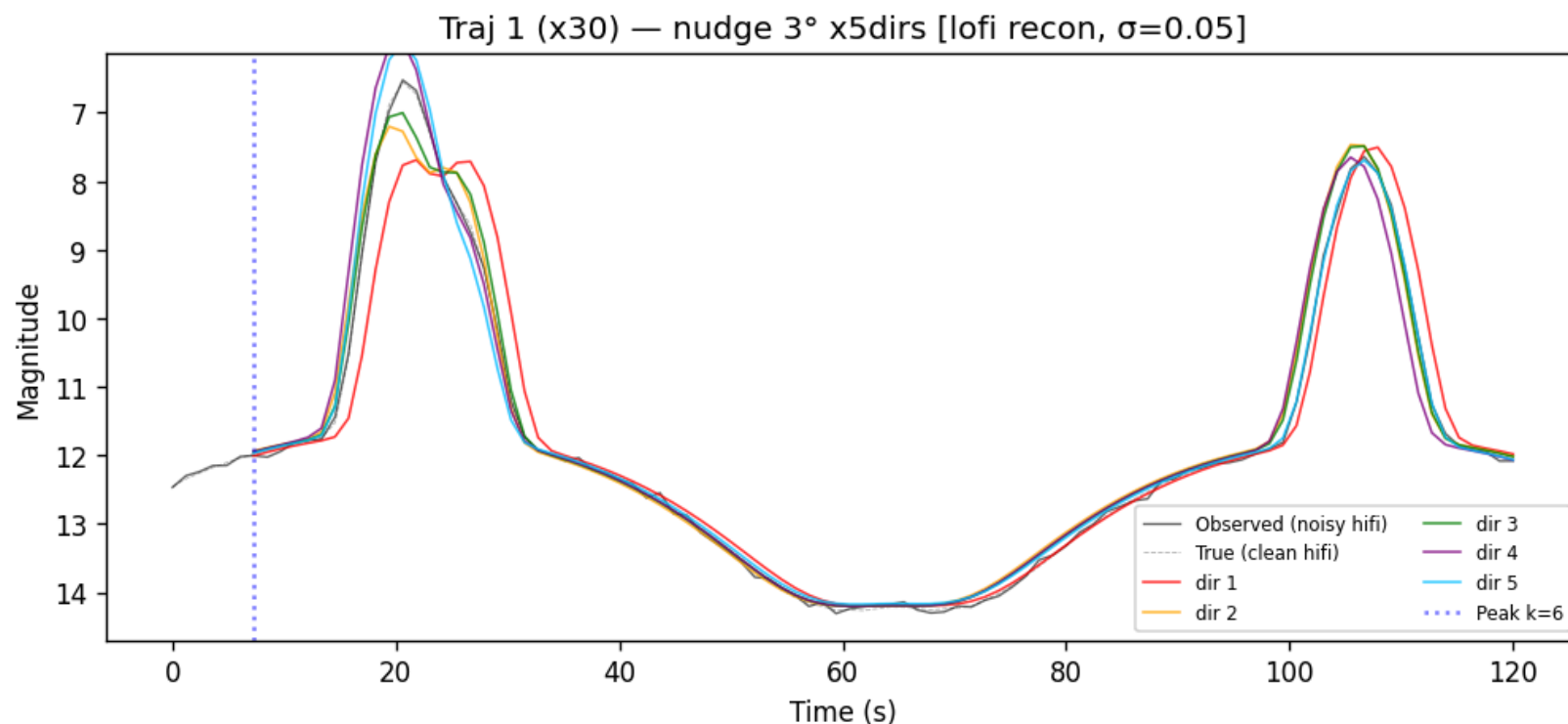


Lo-fi predicts a bright glint at $t \approx 60$ s that does not exist in the hi-fi observations.

This orientation (arbitrary 160° rotation) puts a component in self-shadow at that epoch — hi-fi sees darkness, lo-fi sees a glint. The MSE between lo-fi and hi-fi is large *at truth*, making lo-fi scoring

Lo-fi Failure Is Orientation-Dependent

Same 3° nudge, lo-fi reconstruction, different starting orientation (30° about x):



No spurious peak. Lo-fi and hi-fi agree well. The failure in the previous slide is not a general property of lo-fi — it occurs *specifically* at orientations with active self-shadowing.

Implication: we cannot know a priori which candidate orientations will have lo-fi failures — so lo-fi

What the Nudge Results Mean for the Pipeline

Question	Finding
Can we recover ω cheaply?	Yes — 160 ms, exact with exact endpoints
What if endpoints have 1–5° error?	~0.025 °/s/deg on average, but direction-dependent
Can lo-fi MSE rank candidate pairs?	No — phantom peaks at some orientations make lo-fi MSE unreliable
How accurate do endpoints need to be?	Within ~2° for hi-fi local refinement to converge

Practical consequence: the bridge solver is fine. What we need is (a) accurate endpoint attitudes and (b) hi-fi scoring — never lo-fi — to rank path families.

Staircase: Enumerating the Winding Family

True pair, oracle attitudes, leg spanning 554s. Staircase finds each winding band in turn.

Step	$ \omega $ (°/s)	Arrival err	Trough err
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0	0.316	0.0	1.719
1	0.959	0.0	0.099
2	1.601	0.0	0.021 ← best by trough
3	2.230	0.0	0.792 ← <i>true</i> ($ \omega _{\text{true}} = 2.083^\circ/\text{s}$)
4	2.883	0.036	0.044
5	3.509	0.0	0.134
6	4.180	0.0	1.749
7	4.824	0.0	1.491

8 distinct physically valid bridges. Mechanism confirmed.

Why a Single Trough Fails

Trough scoring selects step 2 (1.601 °/s). **True step is 3 (2.230 °/s).**

The trough epoch is only 202s into a 554s leg. At that early time, steps 2 and 3 produce nearly identical trajectories — the winding difference hasn't accumulated into a detectable brightness signal.

Step 2 scores best by coincidence of timing, not physics. *One checkpoint is not enough.*

Earlier attempt (axis-angle bridge only, same trough): true pair ranked 65/101.

Same root cause — the axis-angle bridge has the wrong winding (step 0), so the trough attitude is completely wrong for every pair including truth.

Angular Momentum Conservation as a Filter

Torque-free dynamics conserves angular momentum:

$$\mathbf{L} = R(q) I \boldsymbol{\omega} = \text{const}$$

Idea: run the staircase independently on *both* legs. At the shared middle peak, only the physically consistent (k, j) pair satisfies $\mathbf{L}_{\text{leg } 0}[k] = \mathbf{L}_{\text{leg } 1}[j]$.

$8 \times 8 = 64$ pairs. One minimum. No extra evaluations needed.

L-Conservation Filter: Result

Ran the filter with oracle attitudes. **Result: incorrect** — identified $(k = 0, j = 0)$, true pair is $(k = 3, j = 1)$.

Root cause: **leg 1 staircase is broken.**

	Leg 0 $ \omega $ (°/s)	Leg 1 $ \omega $ (°/s)
Step 0	0.316	0.249
Step 1	0.959	3.279 ← gap
Step 2	1.601	3.509
Step 3	2.230	3.995

Leg 0 steps uniformly at $\approx +0.65^\circ/\text{s}$ per step. Leg 1 jumps from 0.249 to 3.279, skipping four winding bands entirely. **The L-filter idea is sound — the staircase implementation needs a fix.**

The Near- π Singularity

Leg 1 spans $\Delta t = 720\text{s}$. Its axis-angle starting estimate lands near a π -rad rotation.

The quaternion double-cover introduces a sign ambiguity; the optimiser jumps discontinuously past multiple winding bands.

Fix: initialise step 0 from $\boldsymbol{\omega}_0 = \mathbf{0}$ rather than the axis-angle estimate. Add a tight upper barrier ($\pm 0.4 \delta$ per step) so each solve stays confined to one winding band. The barrier logic is already in the code — one parameter change.

Next Steps

Priority	Action	Goal
1	Fix staircase seed (zero init + tight upper barrier)	Repair leg 1 winding enumeration
2	Rerun L-conservation filter	Confirm unique minimum at true (k, j) with oracle attitudes
3	Test with approximate endpoints	Measure how $\sim 2\text{--}5^\circ$ attitude error degrades L-discrimination
4	Stack multiple troughs	Secondary discriminator if L-filter is insufficient alone

If step 2 succeeds: pipeline becomes — generate candidates/peak → staircase per pair → L cross-match → shortlist correct winding → hi-fi local refinement.